

numpy : is an general purpose library for perform operations on numbers and arrays it provides high performance multidimensionals arrays object with a collection of tools for working efficiently with arrays(matrix)... operation on numpy

1. perform operation in one and multidimensional arrays
2. perform operations of vectors
3. broadcasting capabilities for working with arrays in shapes
4. mathematical , statistical and linear algebra functions are presents

```
import numpy as np
```

arrays is a collection of similar type of element or it is collection of element of the same data , ordered.

```
# create arrays
arr = np.array([1,2,3,4,5])
```

```
print("Arrays : ",arr)
```

```
Arrays :  [1 2 3 4 5]
```

```
print("Shape of arrays : ",arr.shape) # (5,)
print("data type of arrays : ",arr.dtype) #int64
print("number of dim of arrays : ",arr.ndim) #1
print("size of arrays : ",arr.size)
```

```
Shape of arrays : (5,)
data type of arrays : int64
number of dim of arrays : 1
size of arrays : 5
```

```
arr = np.array([
    [1,2,3],
    [4,5,6],
    [7,8,9]
])
print("Shape of arrays : ",arr.shape) # (3,3)
print("data type of arrays : ",arr.dtype) #int64
print("number of dim of arrays : ",arr.ndim) #2
print("size of arrays : ",arr.size)
```

```
Shape of arrays : (3, 3)
data type of arrays : int64
number of dim of arrays : 2
size of arrays : 9
```

```
# access the element from indexing
print(arr[0][0])
```

```
1
```

```
arr = np.zeros((3,4))
print(arr)
```

```
[[0. 0. 0. 0.]
 [0. 0. 0. 0.]
 [0. 0. 0. 0.]
```

```
[0. 0. 0. 0.]
```

```
arr = np.ones((3,4))  
print(arr)
```

```
[[1. 1. 1. 1.]  
 [1. 1. 1. 1.]  
 [1. 1. 1. 1.]]
```

identity matrix

```
arr = np.eye(3)  
print(arr)
```

```
[[1. 0. 0.]  
 [0. 1. 0.]  
 [0. 0. 1.]]
```

range ==> arange(start,end,step)

```
arr = np.arange(0,20,2)  
print(arr)
```

```
[ 0  2  4  6  8 10 12 14 16 18]
```

linspace

```
arr = np.linspace(0,10,5)  
print(arr)
```

```
[ 0.   2.5  5.   7.5 10. ]
```

slicing entire rows

```
arr = np.array([  
    [1,2,3,10],  
    [4,5,6,12],  
    [7,8,9,19]  
])
```

```
print(arr[2,0:2])
```

```
[7 8]
```

Entire columns

```
print(arr[0:,3])
```

```
[10 12 19]
```

